Chapter 16 — Lab Safety and Practical Skills

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 338–339

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Why read the lower meniscus for colourless liquids and the upper meniscus for coloured liquids in a burette?

(a) because it is more convenient

(b) because colorless liquids have more surface tension

(c) because lower meniscus does not exist for colored liquids

(d) because of the parallax effect

Answer: (d) because of the parallax effect — Reading the visible meniscus at eye level avoids parallax error; for coloured liquids the lower meniscus is not clearly seen, so the upper is read.

Option (c) is poorly worded: the lower meniscus exists but is obscured in coloured liquids.

II. Why is phenolphthalein appropriate for strong acid–strong base titrations?

(a) because it is itself weakly acidic

(b) because the pH at equivalence and its colour-change range match

(c) because the solution at the end is acidic

(d) because the solution at the end is basic

Answer: (b) because the pH at equivalence and its colour-change range match — The steep pH change at equivalence (≈7) covers phenolphthalein's colour-change range.

III. Which cation gives a white gelatinous precipitate with aqueous ammonia?

(a) Cr^{3+}

(b) Cr^{2+}

(c) Zn^{2+}

(d) Al^{3+}

Answer: (d) Al^{3+} — Al^{3+} gives white gelatinous $Al(OH)_3$, insoluble in excess NH_3.

IV. NH_4OH gives a green precipitate that turns brown on standing. Which basic radical?

(a) Cu^{2+}

(b) Cr^{3+}

(c) Fe^{2+}

(d) Fe^{3+}

Answer: (c) Fe^{2+} — Fe^{2+} gives green $Fe(OH)_2$ that oxidises to brown $Fe(OH)_3$.

V. Dry heating gives a colourless, odourless gas that turns lime water milky. This suggests:

(a) Chloride ion

(b) Sulfate ion

(c) Carbonate ion

(d) Nitrate ion

Answer: (c) Carbonate ion — CO_2 from carbonate turns lime water milky.

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VI. The chromyl chloride test is a confirmatory test for:

(a) Bromide ions **(b)** Iodide ions
(c) Chloride ions **(d)** Sulfate ions

Answer: (c) Chloride ions — Chromyl chloride (CrO_2Cl_2) test confirms Cl^-.

VII. The brown ring test is a confirmatory test for which acid radical?

(a) Chloride **(b)** Nitrate
(c) Sulfate **(d)** Carbonate

Answer: (b) Nitrate — The brown ring test confirms NO_3^-.

Q.2 — Short Answer Questions

a. For which type of titration is methyl orange used?

Methyl orange is used for strong acid–weak base titrations (equivalence point in the acidic range).

b. Why is phenolphthalein suitable for weak acid–strong base but not strong acid–weak base?

In weak acid–strong base the equivalence point is basic, matching phenolphthalein's range. In strong acid–weak base the equivalence point is acidic, outside phenolphthalein's range, so methyl orange is used instead.

c. Why do different indicators change colour over different pH ranges?

Each indicator is a weak acid/base with its own Ka; the colour change occurs around its pKa, so different indicators change at different pH values.

d. Why is it advisable to use dilute solutions in volumetric analysis?

Dilute solutions give a sharper, more controlled end point and reduce errors from large heat effects or overshooting; they also make the colour change easier to judge.

e. Ca^{2+}, Al^{3+} and Zn^{2+} all give white precipitates with NaOH. How will you detect which is present?

Add excess NaOH: $Al(OH)_3$ and $Zn(OH)_2$ dissolve, $Ca(OH)_2$ does not (or only slightly). Then add excess NH_3: $Zn(OH)_2$ dissolves (deep complex) but $Al(OH)_3$ does not. Ca^{2+} also gives a brick-red flame test.

f. How can Fe^{2+} be distinguished from Fe^{3+} chemically?

Fe^{2+} gives a green precipitate with NaOH (turning brown); Fe^{3+} gives a reddish-brown precipitate. Fe^{3+} also gives a blood-red colour with KSCN, while Fe^{2+} does not.

g. Why does Ca^{2+} not give a precipitate with aqueous ammonia?

Aqueous ammonia provides a low $[OH^-]$; the ionic product does not exceed the Ksp of $Ca(OH)_2$, so no precipitate forms.

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h. How will you find the concentration of acetic acid in vinegar?

Titrate a measured volume of vinegar against standard NaOH using phenolphthalein; use the mole ratio (1:1) and $M_1V_1 = M_2V_2$ to calculate the acetic acid concentration.

i. What precautions are needed while diluting a concentrated acid?

Always add acid slowly to water (never water to acid) with stirring and cooling, because dilution is highly exothermic and may cause splashing.

j. Why does aqueous Na_2CO_3 behave like a base?

CO_3^{2-} hydrolyses: $CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$, producing OH^- and making the solution basic.

k. If aqueous NaOH is kept in an open container, what changes occur over time?

It absorbs CO_2 from air and forms Na_2CO_3 ($2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$), so its concentration decreases and it becomes contaminated with carbonate.

Descriptive / Long Questions

Q.3 Describe common types of Chemistry lab hazards with two examples each.

Physical hazards: broken glass, hot surfaces/electricity. Chemical hazards: corrosive acids/alkalis, toxic/flammable solvents. Biological hazards: microbial cultures, contaminated samples. (Also fire/explosion and radiation hazards where relevant.)

Q.4 What are common accidents in the Chemistry lab? How are they managed in first aid?

Cuts: clean, disinfect, apply pressure for serious bleeding. Acid/alkali splashes: wash with plenty of water (acid then dilute $NaHCO_3$; alkali then dilute boric/acetic acid). Burns: cool and apply suitable dressing. Poisoning: spit/rinse mouth, drink water/milk as appropriate and seek medical help. Fire on clothing: stop, drop and roll/blanket. Eye injury: irrigate with water and seek help.

Q.5 How are the following acid radicals indicated and confirmed: (i) CO_3^{2-} (ii) Cl^- (iii) NO_3^- (iv) SO_4^{2-}?

(i) Carbonate: add dilute HCl → CO_2 turns lime water milky. (ii) Chloride: $AgNO_3$ gives white AgCl soluble in dilute NH_3; chromyl chloride test gives red CrO_2Cl_2 vapours. (iii) Nitrate: brown ring test with $FeSO_4$ and conc. H_2SO_4. (iv) Sulfate: $BaCl_2$ in acidic medium gives white $BaSO_4$ insoluble in acids.

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Q.6 How are the following basic radicals indicated and confirmed: (i) Cu^{2+} (ii) Al^{3+} (iii) Fe^{3+} (iv) Zn^{2+}?

(i) Cu^{2+}: blue precipitate with NaOH; deep blue solution with excess NH_3 ($[Cu(NH_3)_4]^{2+}$). (ii) Al^{3+}: white gelatinous $Al(OH)_3$ soluble in excess NaOH but not NH_3; blue lake test with aluminon. (iii) Fe^{3+}: reddish-brown $Fe(OH)_3$; blood-red with KSCN. (iv) Zn^{2+}: white $Zn(OH)_2$ soluble in excess NaOH and excess NH_3; confirmed with potassium ferrocyanide (white/bluish precipitate).

Quick Check Solutions (in-chapter practice boxes)

Quick Check 16.1 (book p. 330)

a. Mention common types of hazards.

Physical (broken glass, heat, electricity), chemical (corrosive, toxic, flammable, oxidizing), biological (microorganisms), and fire/explosion hazards.

b. How chemical waste is disposed of.

Waste is collected in labelled containers according to its nature (acid/base, organic, heavy-metal, etc.), neutralized/treated where appropriate, and disposed of through approved procedures — not poured randomly into the sink.

c. How an H_2SO_4 burn is treated in the lab.

Wash immediately with plenty of cold running water, then wash with a saturated sodium bicarbonate solution, and rinse again with water; seek medical help for serious burns.

d. Different types of burns.

Dry-heat burns (flame/hot objects), acid burns, alkali burns, and bromine burns.

Quick Check 16.2 (book p. 332)

a. Define indicator and endpoint in a titration.

An indicator is a substance that changes colour at/near the equivalence point. The endpoint is the point at which the indicator changes colour, signalling completion of the titration.

b. What is meant by concordant readings?

Titre readings that agree closely with one another (usually within 0.1 cm^3), showing reproducibility.

c. How the volume of titrant used is calculated.

Volume used = final burette reading − initial burette reading.

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Quick Check 16.3 (book p. 334)

a. Distinguish carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-) radicals.

Both give CO_2 with dilute acid, but carbonates give a white precipitate with $MgSO_4$ in the cold, while bicarbonates do not (or only on boiling). Bicarbonates also give CO_2 on gentle heating of the solid.

b. What is the ring test? Give the equation.

The brown ring test confirms nitrate. Freshly prepared $FeSO_4$ is added to the nitrate solution, then conc. H_2SO_4 is poured down the side of the tube; a brown ring forms at the junction. $NO_3^- + 3Fe^{2+} + 4H^+ \rightarrow NO + 3Fe^{3+} + 2H_2O$; then $[Fe(H_2O)_6]^{2+} + NO \rightarrow [Fe(H_2O)_5NO]^{2+}$ (brown) + H_2O.

c. Names and formulas of some water-insoluble sulphates.

Barium sulphate ($BaSO_4$), lead(II) sulphate ($PbSO_4$), calcium sulphate ($CaSO_4$, sparingly soluble), strontium sulphate ($SrSO_4$).

Quick Check 16.4 (book p. 337)

a. What is the lake test? Give the equation.

The lake test confirms Al^{3+}: aluminium hydroxide adsorbs a dye (e.g. aluminon/blue litmus) to form a coloured "lake". $Al^{3+} + 3OH^- \rightarrow Al(OH)_3\downarrow$; the precipitate adsorbs the dye to give a blue/red lake.

b. How ammonium ion is indicated.

Warm the salt with NaOH solution; NH_3 gas is released, which turns moist red litmus blue and gives white fumes with HCl: $NH_4^+ + OH^- \rightarrow NH_3\uparrow + H_2O$.

c. How to distinguish ferrous (Fe^{2+}) and ferric (Fe^{3+}) radicals.

Fe^{2+} gives a green precipitate with NaOH (turning brown on standing) and a dark blue precipitate with potassium ferricyanide. Fe^{3+} gives a reddish-brown precipitate with NaOH and a blood-red colour with KSCN.

d. What happens when aq. NH_3 is added to Cu^{2+} solution in limited and excess quantities?

Limited NH_3 gives a pale blue precipitate of $Cu(OH)_2$. Excess NH_3 dissolves it to form a deep blue solution of $[Cu(NH_3)_4]^{2+}$.

Textbook checks flagged in this chapter

• p. 338, MCQ I: option (c) is misleading — the lower meniscus exists but is not visible in coloured liquids; the intended answer relates to avoiding parallax/reading the visible meniscus.

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